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The structural Theta method and its predictive performance in the M4-Competition

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ABSTRACT

The Theta method is a well-established prediction benchmark widely used in forecast competitions. This method has received significant attention since it was introduced more than 20 years ago, with several authors proposing variants to improve its performance. This paper considers multiple sources of error versions for Theta, belonging to the family of structural time series models. It investigates its out-of-sample forecast performance using the extensive M4-Competition dataset, which includes 100,000 time series. We compare the proposed structural Theta model against several benchmarks, including all variants of the Theta method. The results demonstrate its remarkable predictive abilities as it outperforms all its variants and competitors, emerging as a solid benchmark for use in forecast competitions.

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1. Introduction

The Theta method was introduced by Assimakopoulos and Nikolopoulos (2000) as a univariate forecasting algorithm that is beneficial for generating predictions of time-series data.[1] It is based on the idea of decomposing a time series into a trend component, representing the long-term pattern of the data, and a residual one, capturing the short-term random fluctuations around the trend. The original Theta method adjusts the local curvature of the series by applying two coefficients, called theta coefficients, to the second differences of the data. The theta coefficients control whether the local curvature is decreased (deflating the series) or increased (dilating the series), thereby amplifying either the long-term or short-term component, respectively (Nikolopoulos & Thomakos, 2019). The resulting two distinct components, referred to as "theta lines", approximate the long and short-term dynamics of the time series, which are subsequently projected and then combined for forecast calculation. This method, as initially proposed by Assimakopoulos and Nikolopoulos (2000), is often referred to as the "classic" Theta method to distinguish it from other variants and subsequent extensions.

Forecasts produced by the classic Theta method are equivalent to those delivered by a simple exponential smoothing with drift, with the restriction that the drift parameter is set to half the slope of the linear trend fitted to the data, see Hyndman and Billah (2003). However, as clearly stated by Nikolopoulos and Thomakos (2019), pp. 11–12, in a standard simple exponential smoothing with drift, both smoothing parameters are typically set or optimised simultaneously in the original time series. Consequently, regardless of the optimisation method, the smoothing parameters coincide with those used in the classic Theta model only by coincidence. For this reason, the forecasts generated by the two models are generally different even though the model's functional form is the same.

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Extensions of the classic Theta method have been put forward by Thomakos and Nikolopoulos (2015), who proposed a multivariate version of the original univariate specification, and by Fiorucci, Pellegrini, Louzada, Petropoulos, and Koehler (2016), who introduced the Dynamic Optimised Theta Model (DOTM). This state-space formulation selects the best short-term theta line and dynamically revises the long-term theta line. More recently, Spiliotis, Assimakopoulos, and Makridakis (2020) suggested three further extensions to the Theta's original framework: considering both linear and non-linear trends, allowing the slope of the trend to be adjusted, and introducing a new multiplicative representation along with the standard additive one.

The classic Theta has been applied in various contexts and exhibited remarkable performance in the M3 competition (Makridakis & Hibon, 2000).[2] As such, it is considered a solid forecast method to beat. The Theta method is also extensively employed by large multi-national organisations (e.g., Uber, Amazon and Bosch) in the private sector. For example, Uber leverages quantitative forecasting methods to predict user supply and demand accurately. Specifically, among statistical models, Uber utilises the Theta approach, which is known for its computational efficiency and demonstrated success in Uber's time series data.[3] Similarly, Amazon Forecast uses the Theta method within the Exponential Smoothing (ETS) built-in algorithm.[4] Additionally, companies like Bosch have adopted this method for predicting demand and sales due to its proven performance and fast implementation. However, as argued by Spiliotis, Assimakopoulos, and Nikolopoulos (2019), despite its good empirical performance, limited academic research has been conducted to generalise its reach and exploit its full potential. This paper aims to address this issue by introducing the structural Theta model, or Multiple Source of Error (MSOE) Theta model, and investigating its prediction performance. The widespread application of the Theta method by leading companies in the private sector underscores the impact of our research.

The structural Theta model examined in this work is an observationally equivalent variant of the single source of error (SSOE) simple exponential smoothing with drift (Hyndman & Billah, 2003). Unlike this latter model, the MSOE version incorporates distinct errors that affect both the states and observations in its state-space formulation (Casals, Sotoca, & Jerez, 2015). In contrast, the simple exponential smoothing with drift relies on a single perturbation term corresponding to the one-step-ahead forecast error to drive the entire system (Hyndman & Billah, 2003).

The structural Theta model can also be connected to the extensive body of literature studying the separation of time series into a long-term trend component and a medium-term business cycle, bearing in mind that the random walk with drift has often been used to estimate the trend (e.g., Beveridge–Nelson decomposition), see Oh, Zivot, and Creal (2008). This literature is closely linked to the broader debate in macroeconomics regarding the significance of permanent and transitory factors in explaining GDP growth (Watson, 1986; Nelson & Plosser, 1982).

Notably, the predictive accuracy of neither the structural Theta model nor the SSOE simple exponential smoothing with drift has been assessed within the framework of the M4 forecast competition, even though the classic Theta has been employed as a benchmark method. This paper effectively fills this gap by explicitly evaluating the forecast accuracy of the structural Theta model using the M4-Competition (Makridakis, Spiliotis, & Assimakopoulos, 2020a), which extends the results of the previous three Makridakis Competitions. The M4-Competition provides a comprehensive database encompassing an extensive collection of time-series data from various industries and economic sectors spanning different time frequencies. The M4-Competition is crucial in enabling more reliable and robust assessments of forecasting methods in an empirical context by including many time series and facilitating the calculation of forecasts.

We decided to focus on the structural Theta model after a close look at the properties of the M4 series. Our analysis reveals that, for most of them, standard unit root tests do not reject the presence of a unit root. Moreover, among the integrated series, a drift is detected in more than one-third of them. These two factors are the crucial ingredients underlying the dynamic properties of the random walk with drift plus noise model, being the state-space representation of our proposed approach. Therefore, the results of this descriptive analysis led us to evaluate the prediction properties of the structural Theta, previously overlooked in the M4-competition.

Forecasts produced by the structural Theta are compared against several competing models, including all benchmarks included in the original M4-Competition, together with Auto Theta (Spiliotis, Assimakopoulos et al., 2020), the Dynamic Optimised Theta Model (Fiorucci et al., 2016) and the Random Walk with Drift plus AutoRegressive model (RWDAR) recently introduced by Sbrana and Silvestrini (2023). Our findings demonstrate the structural Theta's remarkable predictive capabilities, especially for trended series, as it consistently outperforms all its variants and competitors across the 100,000 series in M4. As such, the structural Theta provides a parsimonious yet strong benchmark for business and economic forecasting. The remarkable performance of the structural Theta is further highlighted by the fact that the model can be easily estimated with minimal computational resources and time.

One might be concerned that the excellent performance of the structural Theta is specific to the M4 competition. However, this does not seem to be the case, as the structural Theta also scores well when tested on the time series from the M3 forecasting competition. The forecasting performance of the structural Theta model remains remarkably strong even when tested on the M5-competition dataset, which comprises hierarchical sales

[2] Specifically, it was 3.8% more accurate than the Comb method, see Makridakis, Spiliotis, and Assimakopoulos (2020b).

[3] https://www.uber.com/en-GB/blog/forecasting-introduction/

[4] https://docs.aws.amazon.com/forecast/latest/dg/aws-forecast-recipe-ets.html

data, often characterised by intermittent and erratic patterns. This result is noteworthy given the significant difficulty in predicting such series using conventional forecasting models, such as those used in previous M competitions (Makridakis, Spiliotis, & Assimakopoulos, 2022).

The rest of the paper proceeds as follows. Section 2 presents the structural Theta model and its estimation procedure. Section 3 focuses on some time-series properties (presence of drift and persistence) of the M4-dataset. Section 4 illustrates a Monte Carlo experiment designed for comparing the forecast accuracy of the structural Theta model relative to other benchmark models on simulated data, controlling for the presence of drift and persistence components. Section 5 evaluates the forecast accuracy of the structural Theta model on the M4-Competition dataset. Section 6 examines the performance of the structural Theta using the M3-Competition data. Section 7 discusses a potential implementation of the structural Theta on the M5-Competition dataset. Section 8 provides some final comments.

2. Model and estimation approach

2.1. The structural theta model

The state-space representation of the structural Theta model is as follows:

$$\begin{aligned} y_t &= \mu_{t-1} + \epsilon_t \qquad & \epsilon_t &\sim \mathcal{NID}(0, \sigma_\epsilon^2) \\ \mu_t &= \omega + \mu_{t-1} + \eta_t & \eta_t &\sim \mathcal{NID}(0, \sigma_\eta^2) \end{aligned} \tag{1}$$

where y_t is the observed time series, μ_t is its level (or stochastic trend), ω is a constant term, while $\mathcal{NID}$ denotes normally and independently distributed innovations and $t = 1, 2, \ldots, n$ is the number of observations. It is also assumed that $E(\epsilon_t \eta_t) = 0$. In contrast to the single source of error state-space approach, the structural Theta model in (1) is driven by two uncorrelated noises, resulting in a formulation known as an MSOE, and belongs to the class of "unobserved components time series models", or "structural time series models", pioneered by Harvey (1989).

Model (1) has an ARIMA(0,1,1) with drift reduced form:

$$(1 - L)y_t = \omega + (1 - L)\epsilon_t + \eta_{t-1} = \omega + v_t + \theta v_{t-1} \tag{2}$$

where L is the lag operator, such that $Ly_t = y_{t-1}$. The second equality in (2) holds true as far as the two expressions are observationally equivalent, as they share the same autocorrelation structure (i.e., an autocorrelation function, which is zero after the first lag). Indeed, the sum $(1 - L)\epsilon_t + \eta_{t-1}$ is a first-order moving average process, which can be rewritten as $v_t + \theta v_{t-1}$, where θ is the moving average coefficient and v_t is the innovation term. The corresponding forecast function is equal to a simple random walk update plus a constant linear trend: $y_{t+h} = \omega h + y_t \quad (\forall h > 0)$, see Hyndman and Billah (2003), p. 289. In the empirical context, Nelson and Plosser (1982) and Rossana and Seater (1995) provide substantial evidence supporting the applicability of the ARIMA(0,1,1) model to a variety of macroeconomic time series. Additionally, Working (1960) demonstrates that temporal aggregation of a high-frequency random walk leads to an ARIMA(0,1,1) model with a positive-valued moving average coefficient.

To shed further light on the connection between the structural Theta and the classic Theta model, it should be noted that Hyndman and Billah (2003) showed that, under certain parameter restrictions, the underlying stochastic model of the classic Theta method, as proposed by Assimakopoulos and Nikolopoulos (2000), is a simple exponential smoothing with drift, which is also equivalent to an ARIMA(0,1,1) model with drift. Consequently, the structural Theta model introduced in this paper and the classic Theta method share the same ARIMA(0,1,1) reduced form. This elucidates why Eq. (1) is an equivalent representation of the original Theta model.

Once fed into its state-space representation, model (1) can be estimated by the Kalman filter. The structural Theta features three parameters; however, as shown by Harvey (1989), the model can be concentrated out by considering the "signal-to-noise" ratio: $q_\eta = \frac{\sigma_\eta^2}{\sigma_\epsilon^2}$. This reduces the number of parameters to be estimated and simplifies the Kalman filter recursions as follows (see Harvey, 1989, p. 107, Example 3.2.1):

- **Innovation** : $v_t = y_t - m_{t-1}$
- **Innovation variance** : $f_t = p_{t-1} + 1$
- **Kalman gain** : $k_t = \frac{p_{t-1}}{p_{t-1}+1}$
- **Prediction mean** : $m_t = E(\mu_t | Y_{t-1}) = \omega + m_{t-1} + k_t v_t$
- **Prediction variance** : $p_t = E(\mu_t - m_t)^2 = p_{t-1} - \frac{p_{t-1}^2}{p_{t-1}+1} + q_\eta$

where $Y_{t-1} = \{y_1, y_2, \ldots, y_{t-1}\}$ denotes the information provided by data up to time $t-1$ and $Var(v_t | Y_{t-1}) = \sigma_\epsilon^2 f_t$.

These expressions can derive the prediction intervals (PI) for y_t at time $n + h$, $h = 1, 2, \ldots$. Those are:

$$PI(\hat{y}_{n+h|n}) = \hat{y}_{n+h|n} \pm z_{(1-\alpha)} \sqrt{MSE(\hat{y}_{n+h|n})} \tag{3}$$

where $\hat{y}_{n+h|n}$ is the h-step-ahead prediction of y_{n+h}, $z_{(1-\alpha)}$ is the quantile of a standard normal distribution at the desired significance level α and the forecast Mean Squared Error (MSE) is

$$MSE(\hat{y}_{n+h|n}) = \sigma_\epsilon^2 (p_n + 1 + (h-1)q_\eta)$$

see Harvey (1989), p. 148.

The model in (1) can be easily estimated by selecting the two parameters (q_η and ω) that maximise the *concentrated* log-likelihood function: $\log L_c = -\frac{n}{2}\log(2\pi + 1) - \frac{1}{2}\sum_{t=1}^{n}\log(f_t) - \frac{n}{2}\log(\frac{1}{n}\sum_{t=1}^{n}\frac{v_t^2}{f_t})$ (see Harvey, 1989, p. 127, equation (3.4.9a)). Equivalently, one can select the two parameters that minimize $\sum_{t=1}^{n}\log(f_t) + n\log(\frac{1}{n}\sum_{t=1}^{n}\frac{v_t^2}{f_t})$.

The estimation comes at very low computational costs and no convergence issues.[5] This aspect is a significant benefit of the structural Theta model, particularly when compared to more sophisticated and computationally intensive methods.

[5] The R code containing the estimation and prediction using the structural Theta method is available at the following link: https://rpubs.com/giac76/1167511

Table 1
Data used for the empirical evaluation (M4 dataset).

Frequency	Number of series	Forecast horizon
Yearly	23,000	6
Quarterly	24,000	8
Monthly	48,000	18
Weekly	359	13
Daily	4227	14
Hourly	414	48

2.2. Seasonality in the structural Theta model

The model in (1) does not incorporate seasonality explicitly and consequently depends on an "external" seasonal adjustment procedure, in which the data are seasonally adjusted before conducting estimation and forecasting. Therefore, we employ the conventional multiplicative decomposition approach when deemed necessary based on the results of a standard seasonality test. The same approach is used for the classic Theta method in the M4-Competition, see Table 2 in Makridakis et al. (2020a).

That is, defining s the seasons ($s \neq 1$), and assuming s is even, the smoothed series is given by $y_t^* = \sum_{j=-\frac{s}{2}}^{\frac{s}{2}} w_j y_{t-j}$ with $t = \frac{s}{2}+1, \ldots, n-\frac{s}{2}-1$ where $w_j = \frac{1}{s}$ for $j = 0, \pm 1, \ldots, \pm(\frac{s}{2}-1)$ and $w_{-j} = w_j = \frac{1}{2s}$ for $j = \pm\frac{s}{2}$. The ratios $\frac{y_t}{y_t^*}$ are then averaged over each season to give a set of seasonal factors. Dividing the original series by these seasonal factors results in the seasonally adjusted series. The structural Theta forecasts carried out on the deseasonalised series are finally multiplied by the seasonal factors to obtain the seasonal forecasts.

3. The M4 competition dataset

3.1. Overview

Forecast competitions have greatly influenced the field of forecasting over the years, providing a solid basis for assessing different approaches and learning how to advance forecasting theory and practice (Makridakis et al., 2020a).

The accuracy of the structural Theta model is tested on the M4-Competition dataset, which provides a comprehensive collection of time-series data from various domains, including finance, economics, demographics, and industry. In addition, the dataset covers a broad range of frequencies, such as yearly, quarterly, monthly, weekly, and daily, along with hourly data, allowing for the assessment of forecasting methods across different time horizons.

Table 1 provides several insights into the M4 dataset, offering details on various aspects, such as the series distribution across the different frequencies and the forecasting horizons per data frequency. The yearly, monthly, and quarterly series collectively comprise the vast majority of the dataset (95,000-time series), indicating their substantial influence on the overall outcomes of the forecast competition.

Each time series in the dataset is partitioned into a training set, used to estimate the different competing models, and a test/validation set employed to calculate forecast errors and assess the accuracy of the models.

3.2. Persistence and trend significance in M4 series

The *ex-ante* evaluation of time-series properties is crucial for formulating a reliable forecasting model. Indeed, it ensures that the modelling approach aligns with the data characteristics. In this section, we focus primarily on two time-series properties of the M4 dataset, namely, persistence and trend significance, see Spiliotis, Kouloumos, Assimakopoulos, and Makridakis (2020). We claim that understanding these properties can help interpret the results of the forecast competition because the structural Theta model is well-suited for handling unit root data-generating processes. Although understanding the main time-series characteristics of such a large dataset as M4 is challenging, incorporating them into the forecasting framework is the key to successful forecasting.

The autoregressive (AR) process provides a clear, intuitive measure of persistence (Spiliotis, Kouloumos et al., 2020). Andrews and Chen (1994) suggest using the sum of the AR coefficients to assess the degree of persistence of a time series based on the monotonic relation between the sum of the AR coefficients and the cumulative impulse response, which is a useful summary of the information contained in the impulse response function. Fig. 1 thus illustrates the empirical distribution of the sum of the coefficients of an AR(p) process estimated on all the M4-Competition series in levels,[6] previously seasonally adjusted if deemed necessary based on the output of a standard seasonality test. Each chart refers to a single frequency (yearly, quarterly, monthly and daily) and shows the histogram with density values on the left y-axis. The corresponding empirical cumulative distribution function is also added on the right y-axis to provide an alternative visualisation of the sample distribution.

Focusing on yearly data shows that the empirical distribution of the sum of the AR coefficients is left-skewed, with most values falling within the range [0.8, 1]. The same observation holds for the quarterly, monthly and daily frequencies, which tend to have a higher concentration of values in this interval.

Even at weekly and hourly frequencies, the distribution of the sum of the autoregressive coefficients is concentrated around one (Fig. 2); yet, a non-negligible part of the probability distribution lies in the [0, 0.8] interval.

Overall, as highlighted by Castle, Doornik, and Hendry (2021), it is clear that the estimates of the sum of the AR(p) coefficients cluster near unity most of the time, pointing to high persistence in the data or near-unit root behaviour.

A similar and stronger message arises when conducting unit root tests on the M4-Competition time series. Specifically, Table 2 (top part) presents the results of augmented Dickey–Fuller tests, see Dickey and Fuller (1979), performed across all series for each time-frequency, using

[6] The maximum order of the AR(p) process is set equal to the frequency of the data; for example, the maximum p is equal to 4 for quarterly data and to 12 for monthly data. The analysis has also been performed assuming an AR(1) process instead of an AR(p), resulting in similar charts across all time frequencies, which the authors can provide upon request.

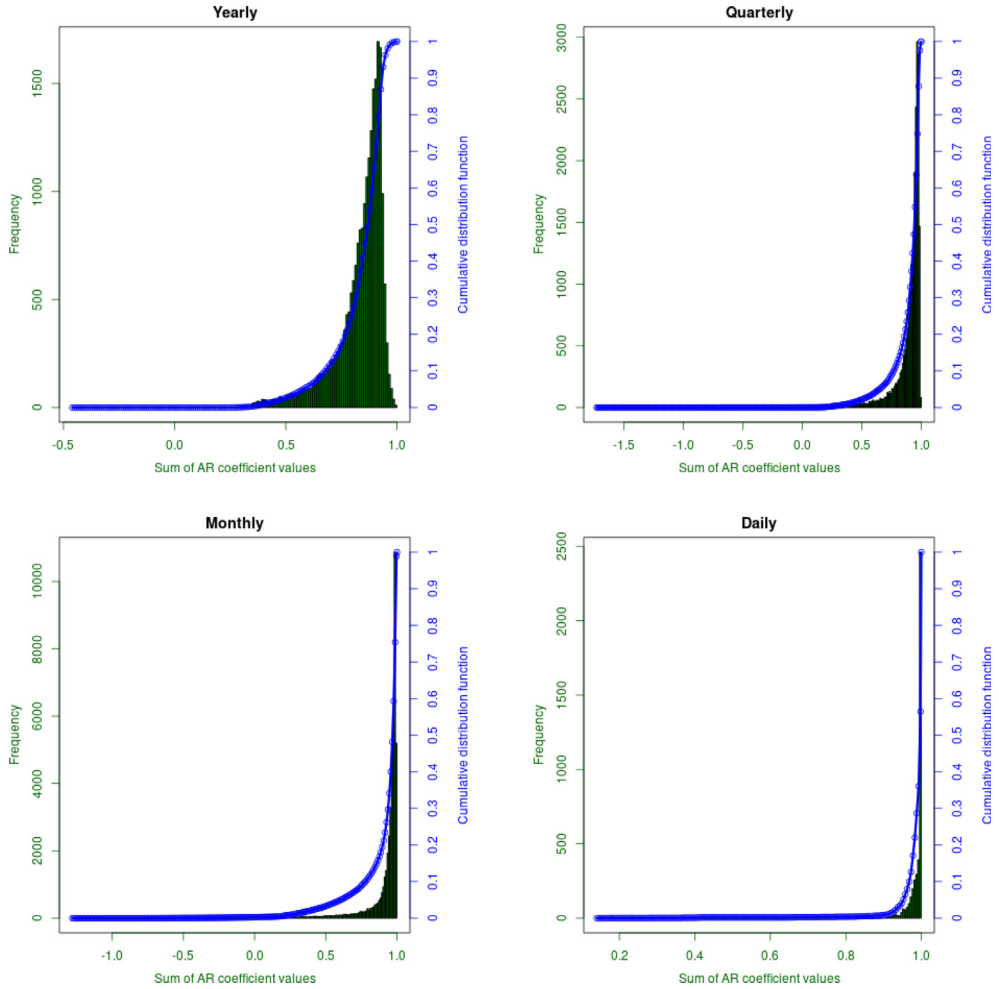


Fig. 1. Distribution of the sum of the AR(p) coefficients estimated on the M4-Competition series across yearly, quarterly, monthly and daily frequencies.
Notes. This figure shows the histograms and the empirical distribution functions of the sum of the AR(p) coefficients estimated on the time series of the M4-Competition across different frequencies (yearly, quarterly, monthly and daily).

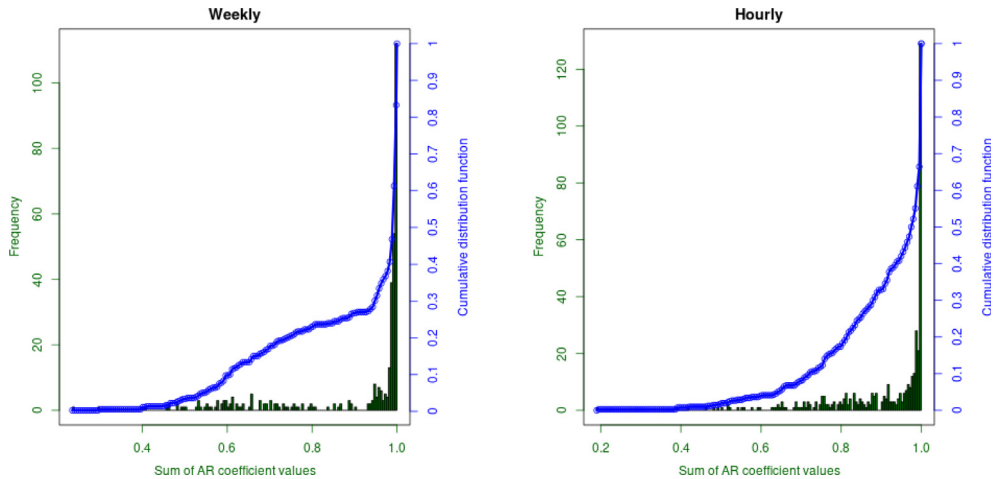


Fig. 2. Distribution of the sum of the AR(p) coefficients estimated on the M4-Competition series across weekly and hourly frequencies.
Notes. This figure shows the histograms and the empirical distribution functions of the sum of the AR(p) coefficients estimated on the time series of the M4-Competition across the remaining frequencies (weekly and hourly).

Table 2
Percentage of series with a unit root and with a drift (given a unit root) across the M4-Competition dataset.

Frequency	% series with a unit root
Yearly	94.857
Quarterly	88.892
Monthly	76.467
Daily	95.481
Weekly	64.067
Hourly	32.126
Frequency	**% series with a drift given a unit root**
Yearly	51.581
Quarterly	45.631
Monthly	33.801
Daily	19.530
Weekly	36.271
Hourly	21.452

Notes. The top part of the table reports the percentage of series for which the null hypothesis of a unit root is not rejected at the 5% significance level, based on the output of an augmented Dickey–Fuller test; see Dickey and Fuller (1979) and Dickey and Fuller (1981). The bottom part of the table presents the percentage of series for which the null hypothesis of no drift given a unit root is rejected at the 5% significance level, based on the same test.

a standard significance level of 5%. The test's null hypothesis assesses whether the time series has a unit root instead of a stationary alternative hypothesis. The test can be executed in three versions: the first with no drift and no linear trend, the second with drift but no linear trend, and the third with both drift and a linear trend. The results in the top part of Table 2 pertain to the second version.

The findings reveal that a significant portion of the series exhibits a unit root, particularly at yearly, quarterly, monthly, and daily frequencies. The percentage of first-order integrated time series is also notably high at the weekly frequency. However, for hourly series, this proportion is closer to one-third. Overall, it is observed that more than 80% of the 100,000 time series in the M4-Competition dataset contain a unit root at the 5% significance level.

A closely related issue concerns the presence of a drift in the time series. Again, we consider the standard augmented Dickey–Fuller equation with an intercept added to the test regression. Then, for those series for which the null hypothesis of a unit root is not rejected, it is possible to test the significance of the drift term given that the unit root is present, see Enders (1995) p. 223 (Table 4.1). The outcome of these t-tests ($\tau_{\alpha\mu}$) is summarised in the bottom part of Table 2.

Given a unit root, the percentage of series with a drift is quite high at the yearly and quarterly frequencies, where about half of those with a unit root also have a drift. This percentage drops to one-third for the monthly series. Overall, however, it emerges that more than 40% of the integrated series also have a drift at the 5% significance level. This share is anything but marginal. This evidence represents the starting point that led us to investigate further the prediction performance of the structural Theta model in the framework of the M4 competition.

4. Monte Carlo study

This section presents a Monte Carlo simulation that evaluates the forecasting performance of the structural Theta model while controlling for the presence of deterministic drift and persistence in the data-generating process (henceforth DGP).[7]

In this Monte Carlo experiment, data are simulated assuming either a trend stationary process (DGP 1) or a difference stationary process (DGP 2), following the seminal work of Nelson and Plosser (1982). These two classes of non-stationary processes imply different dynamics and forecasts, thereby allowing us to validate the hypothesis that the structural Theta performs better in the presence of persistence and deterministic drift components in the time series (see Box I):

In DGP 1, α and β are fixed parameters, while c_t follows a stationary ARMA(1,1) process. DGP 1 exhibits a deterministic trend over time, but the fluctuations around this trend are stationary; the trend component can be removed through differencing, resulting in a stationary residual series. The variance of the series remains constant over time once the trend is removed.

In DGP 2, y_t is taken in first differences, and d_t is a stationary ARMA(1,1). Therefore, DGP 2 accumulates an ARMA(1,1) over time, resulting in a degree of persistence higher than that of DGP 1.[8] Differencing y_t removes the trend and other deterministic components, rendering the series stationary. Unlike DGP 1, DGP 2 may exhibit time-varying variance.

The sample size is 60 observations, and the number of replications is 10,000. All noises follow a Gaussian distribution. Data are simulated assuming random parameters. That is, for each replication, α, β and all the autoregressive and moving average coefficients are independently drawn from Uniform distributions.[9] The use of random – rather than fixed – parameters allows the simulation experiment to be general rather than case-specific.

For each simulated series, six out-of-sample values are used as the hold-out period to compare the forecast performance of the structural Theta model to ETS and auto ARIMA (Hyndman & Khandakar, 2008), two standard statistical benchmark methods also used in the original M4-Competition. Both point forecasts and prediction intervals are considered.

[7] The R code developed to reproduce the results in this section and the following sections is available at these two links: https://rpubs.com/giac76/A1198174 and https://rpubs.com/giac76/1198250.

[8] DGP 2 can also be cast in the state-space form:

$$\begin{aligned} y_t &= \mu_t + \chi_t \\ \mu_t &= \mu_{t-1} + \beta \\ \chi_t &= \chi_{t-1} + d_t \end{aligned} \tag{5}$$

and d_t follows a stationary ARMA(1,1) process.

[9] Let us denote with $Uniform(a, b)$ a Uniform distribution defined over the interval a and b. In DGP 1, the AR parameter is drawn from a $Uniform(0.7, 0.99)$ and the MA parameter from a $Uniform(-0.7, -0.1)$, while α and β from a $Uniform(0.1, 5)$ and a $Uniform(0.01, 0.1)$, respectively. In DGP 2, the AR parameter is drawn from a $Uniform(0.1, 0.7)$ and the MA parameter from a $Uniform(-0.7, 0)$, while the intercept from a $Uniform(0.01, 0.1)$.

$$\begin{array}{lll} \textit{DGP 1} & \textit{DGP 2} & \\ y_t = \alpha + \beta t + c_t & (1-L)y_t = \beta + d_t & t = 1, 2, \ldots, n \\ (1-\phi L)c_t = (1+\theta L)u_t & (1-\delta L)d_t = (1+\lambda L)u_t & u_t \sim \mathcal{NID}(\mu_u, \sigma_u^2) \end{array} \tag{4}$$

Box I.

$$MSIS = \frac{1}{h} \frac{\sum_{t=n+1}^{n+h}(u_t - l_t) + \frac{2}{a}(l_t - y_t)\mathbb{1}\{y_t < l_t\} + \frac{2}{a}(y_t - u_t)\mathbb{1}\{y_t > u_t\}}{\frac{1}{n-s}\sum_{t=s+1}^{n}|y_t - y_{t-s}|}$$

Box II.

Table 3
Monte Carlo averages of RMSSEs for point forecasts and MSISs for prediction intervals constructed using the competing methods.

	DGP 1			DGP 2		
	Root Mean Squared Scaled Error (RMSSE)					
	MSOE Theta	ETS	AutoARIMA	MSOE Theta	ETS	AutoARIMA
h = 1	**0.716**	0.728	0.729	0.778	0.776	**0.771**
h = 2	**0.840**	0.856	0.857	**1.033**	1.040	1.038
h = 3	**0.914**	0.935	0.933	**1.234**	1.253	1.253
h = 4	**0.972**	0.999	0.994	**1.404**	1.438	1.437
h = 5	**1.020**	1.054	1.046	**1.563**	1.613	1.612
h = 6	**1.062**	1.103	1.090	**1.709**	1.779	1.776
	Mean Scaled Interval Score (MSIS)					
	MSOE Theta	ETS	AutoARIMA	MSOE Theta	ETS	AutoARIMA
85%	**5.554**	5.957	5.728	**9.465**	9.983	9.661
95%	**7.115**	7.569	7.293	12.686	12.595	**12.593**

Notes. The top part of this table refers to point forecasts, while the bottom refers to prediction intervals. The benchmark methods used for comparison purposes are ETS and auto ARIMA. The most accurate method for RMSSE/MSIS is highlighted in boldface.

Results are presented in Table 3, with the top part showing point forecasts and the bottom part showing prediction intervals. Point forecast are compared using the Root Mean Squared Scaled Error (RMSSE):

$$RMSSE = \sqrt{\frac{\frac{1}{h}\sum_{t=n+1}^{n+h}(y_t - \hat{y}_t)^2}{\frac{1}{n-s}\sum_{t=s+1}^{n}(y_t - y_{t-s})^2}}$$

where y_t are the future observations of the series, $\hat{y}_t$ is the forecast made at time t, h is the forecasting horizon and s the frequency of the data (in this Monte Carlo $s = 1$).

Prediction intervals are evaluated using the Mean Scaled Interval Score (MSIS) metrics as follows (see Box II): where l and u represent the lower and upper bounds of the prediction intervals, a is the significance level (set to 0.05), and $\mathbb{1}$ is the indicator function.

Regarding point forecasts, the structural Theta model tends to outperform the other two competitors across all forecast horizons (except for $h = 1$ with DGP 2, when the three models have an almost equal performance). Regarding prediction intervals, the structural Theta is more accurate than ETS and auto ARIMA, except for DGP 2 at 95%. These findings confirm that the structural Theta generally outperforms standard and flexible benchmarks such as ETS and auto ARIMA whenever data are generated by random walks or by stationary processes fluctuating around a deterministic trend.

5. Empirical evaluation using the M4 Competition data

The previous section has shown that the structural Theta model outperforms standard statistical benchmarks when using simulated data with varying degrees of persistence and drift components. We now turn to the analysis of real-world time series and compare the performance of the structural Theta with the main benchmark models used in the M4-competition dataset. These benchmarks encompass SES, Holt, Damped, and the Comb model, corresponding to the simple arithmetic average of the first three models. This was used for comparing all of the submitted methods; see Table 2 on page 57 in Makridakis et al. (2020a). Furthermore, besides the classic Theta method, several alternative versions and extensions of it are considered, like Auto Theta (Spiliotis, Assimakopoulos et al., 2020), the Dynamic Optimized Theta Model (Fiorucci et al., 2016), and the Random Walk with Drift plus AutoRegressive model (RWDAR), see Sbrana and Silvestrini (2023).

The focus is, initially, on point forecasts. We assess the prediction performance using the same metrics adopted for the M4-competition (see Makridakis et al., 2020a),

that is, the symmetric mean absolute percentage error (or sMAPE, introduced by Makridakis, 1993), the mean absolute scaled error (or MASE) and the Overall Weighted Average (OWA), which corresponds to the average of the previous two:

$$MASE = \frac{1}{h}\frac{\sum_{t=n+1}^{n+h}|y_t-\hat{y}_t|}{\frac{1}{n-s}\sum_{t=s+1}^{n}|y_t-y_{t-s}|}$$
$$sMAPE = \frac{2}{h}\sum_{t=n+1}^{n+h}\frac{|y_t-\hat{y}_t|}{|y_t|+|\hat{y}_t|}\times 100, \quad (6)$$

where y_t is the observed value of the series at time t, $\hat{y}_t$ is the forecast made at time t, h is the forecast horizon, n is the number of data points in the in-sample, while s is the frequency of the data.

5.1. Results for the yearly, quarterly, monthly and daily series

Table 4 presents the yearly, quarterly, monthly, and daily series results. At seasonal frequencies, models are estimated on the seasonally adjusted data whenever necessary based on a standard seasonality test output. Then, the final forecasts are reseasonalised following the procedure outlined in Section 2.2. Each column refers to a model, while each row reports the sMAPE, MASE, and OWA values for evaluating the model's forecast accuracy. The overall ranking obtained by applying the OWA metric (calculated by expressing a model's sMAPE and MASE as a ratio of sMAPE and MASE achieved by the naive/random walk forecast) is also presented. The best result in terms of OWA is marked in bold.

Focusing on the yearly data, the structural Theta stands up as the most accurate model, followed in the second position by the RWDAR (Sbrana & Silvestrini, 2023), which is akin to a random walk with drift plus an autoregressive term, and therefore shares many similarities with the model (1). Unreported results show that the structural Theta model exhibits very similar predictive performance to the SSOE Theta variant, as largely anticipated, given their observational equivalence. This finding holds true in general also for the other time frequencies, not only for annual data.

The two extensions of the classic Theta (DOTM and Auto Theta) are positioned in the third and fourth rankings. The DOTM performs similarly to RWDAR, while Auto Theta lags far behind. The Comb model ranks fifth, and the classic Theta method follows immediately after, with an OWA equal to 0.872 (9% higher than the structural Theta). All the other models/methods (SES, Holt and Damped) exhibit significantly lower performance.

The quarterly dataset conveys a similar message. The structural Theta model yields the lowest OWA, followed by DOTM and Auto Theta at a greater distance, and then by the Comb method, which holds the third position. The structural Theta seems superior in forecast accuracy for the yearly and quarterly series compared to the Theta classic.

In the monthly data, the ranking is slightly different since the structural Theta model comes in the third position (0.916), anticipated by Auto Theta and the Theta classic. These results align closely with Table 2. Indeed, while for yearly and quarterly frequencies, we observe a strong presence of persistence and drift (nearly one series out of two), for monthly series, we detect the same in only about one series out of three. However, despite not being the top-performing model, the structural Theta model still displays strong forecasting capabilities when applied to monthly time series.

Moreover, our proposed approach ranks second using daily series, outperforming all variants of Theta.[10]

5.2. Weekly and hourly time series

On the other hand, for both hourly and weekly series, it is evident that the performance of the structural Theta model deteriorates, as shown in Table 5. This finding aligns with the properties described in Section 3.2, with particular reference to the integration order and presence of trend.

5.3. Overall performance

We now evaluate the average performance of the models across all the M4 time series. Table 6 shows that the structural Theta model outperforms all competitors in our forecasting exercise. Additionally, the structural Theta model performs very well even among all the 61 methods considered for point forecasts in the M4-Competition.[11] Indeed, in terms of OWA, it ranks in the 10th position (we refer to Table 4 in Makridakis et al., 2020a). Noteworthy, the structural Theta model outperforms all benchmarks (classic Theta, Comb, Damped, Holt, SES, Naive 2, Naive 1, Naive S, RNN, MLP) and two standard methods (ARIMA, ETS). It also beats sophisticated extensions of the classic Theta method, such as Auto Theta and DOTM. We believe that these points emphasise the significance of our contribution.

How can such a simple and parsimonious model consistently outperform many statistically sophisticated competitors, even ranking among the top performers? We believe the answer lies in the descriptive analysis in Section 3.2, which lays the groundwork for the subsequent forecast evaluation. As observed, the prevalence of unit roots and drifts in the series strongly supports our simple forecasting approach. Indeed, these two factors seem to be particularly relevant for successful out-of-sample forecasting.

5.4. Multiple comparisons with the best (MCB)

In this section, we implement the nonparametric Nemenyi test (Nemenyi, 1963), which is comparable to the multiple-comparisons procedure described by Koning, Franses, Hibon, and Stekler (2005) (multiple comparisons with the best method, MCB). This test assesses the statistical significance of observed accuracy differences between

[10] With the Dynamic Optimized Theta Model (DOTM), three daily series have been excluded (2047, 2193, 4097) due to some estimation issues.

[11] These 61 competitors include 49 submitted methods, ten benchmarks, and two standard methods for comparison.

Table 4
Forecasting performance in terms of point forecasts across the Yearly, Quarterly, Monthly and Daily M4 series.

Methods	MSOE Theta	SES	Holt	Damped	Comb	Theta classic	DOTM	Auto Theta	RWDAR
	Yearly								
sMAPE	**13.507**	16.398	16.535	15.163	14.874	14.603	13.677	13.797	13.690
MASE	**3.054**	3.981	3.576	3.372	3.282	3.382	3.075	3.184	3.059
OWA	**0.798**	1.003	0.956	0.888	0.868	0.872	0.805	0.823	0.804
Rank (OWA)	**1**	9	8	7	5	6	3	4	2
	Quarterly								
sMAPE	**10.023**	10.600	10.955	10.243	10.197	10.312	10.089	10.125	10.312
MASE	**1.165**	1.340	1.199	1.175	1.174	1.232	1.185	1.179	1.181
OWA	**0.880**	0.970	0.935	0.893	0.891	0.917	0.890	0.890	0.899
Rank (OWA)	**1**	8	7	4	3	6	2	2	5
	Monthly								
sMAPE	13.271	13.618	14.828	13.473	13.434	13.003	13.321	**13.125**	13.368
MASE	0.969	1.020	1.010	0.972	0.966	0.970	0.977	**0.960**	0.969
OWA	0.916	0.951	0.989	0.924	0.920	0.907	0.921	**0.906**	0.919
Rank (OWA)	3	8	9	7	5	2	6	**1**	4
	Daily								
sMAPE	3.014	3.045	3.070	3.063	**2.985**	3.053	3.042	3.025	3.064
MASE	3.226	3.281	3.231	3.234	**3.205**	3.262	3.209	3.231	3.265
OWA	0.987	1.000	0.997	0.996	**0.979**	0.999	0.997	0.990	1.001
Rank (OWA)	2	7	5	4	**1**	6	5	3	8

Notes. The results are presented per data frequency. The most accurate method in terms of OWA is highlighted in boldface.

Table 5
Forecasting performance in terms of point forecasts across the Weekly and Hourly M4 series.

Methods	MSOE Theta	SES	Holt	Damped	Comb	Theta classic	DOTM	Auto Theta	RWDAR
	Weekly								
sMAPE	8.939	9.012	9.706	**8.867**	8.947	9.094	9.089	8.987	8.548
MASE	2.547	2.685	2.413	**2.400**	2.429	2.637	2.583	2.488	2.543
OWA	0.946	0.975	0.964	**0.916**	0.926	0.971	0.961	0.938	0.924
Rank (OWA)	5	9	7	**1**	3	8	6	4	2
	Hourly								
sMAPE	18.661	18.094	29.474	19.277	22.114	18.138	18.093	17.754	**14.093**
MASE	3.418	2.385	9.380	2.947	4.585	2.455	2.388	1.807	**2.253**
OWA	1.221	0.990	2.760	1.140	1.559	1.006	0.991	0.860	**0.854**
Rank (OWA)	7	3	9	6	8	5	4	2	**1**

Notes. The results are presented per data frequency. The most accurate method in terms of OWA is highlighted in boldface.

Table 6
Forecasting performance in terms of point forecasts across all the M4-Competition series.

Methods	MSOE Theta	SES	Holt	Damped	Comb	Theta classic	DOTM	Auto Theta	RWDAR
	All frequencies								
sMAPE	**12.120**	13.088	13.836	12.654	12.566	12.312	12.197	12.137	12.258
MASE	**1.607**	1.885	1.779	1.682	1.664	1.697	1.615	1.629	1.609
OWA	**0.867**	0.975	0.975	0.906	0.898	0.898	0.872	0.873	0.873
Rank (OWA)	**1**	6	6	5	4	4	2	3	3

Notes. The most accurate method in terms of OWA is highlighted in boldface.

forecasts; see Demšar (2006) and Hollander, Wolfe, and Chicken (2013). It ranks the performance of methods for each time series, computes the means of those ranks (medians), and produces confidence intervals for those means. If the confidence intervals for different methods overlap, it indicates that the means are not significantly

different. Otherwise, it identifies which method has a higher rank and which has a lower one.

The Nemenyi test is conducted separately for yearly, quarterly, monthly, and daily series from the M4 Competition dataset, employing the Mean Absolute Scaled Error (MASE) metric as used by Spiliotis, Assimakopoulos et al. (2020), and the `nemenyi` function in the `tsutils` package for R. The results of the Nemenyi test are presented in Fig. 3 for each time frequency.

Fig. 3 shows that the Friedman test rejects the null hypothesis of no difference in accuracy among the forecast models at all frequencies. Thus, post-hoc analysis based on the Nemenyi test is performed to evaluate which differences are significant. The structural Theta model exhibits the best-ranked performance for yearly data, which is not statistically different from that of RWDAR, DOTM, Auto Theta, and Comb. However, the structural Theta model statistically outperforms all other methods above the reference value without any overlap with the shaded area. A similar ranking emerges for quarterly data, with the structural Theta model ranked first along with Comb. The structural Theta is no longer the best model for monthly data, but its performance is not statistically different from the best forecasting method (Auto Theta). Even with daily data, the structural Theta ranks third and consistently outperforms classic Theta, DOTM, RWDAR, and SES.

Drawing overall conclusions from this analysis, the previous results described in Section 5.1 are broadly confirmed: the structural Theta model is found to perform well compared to other benchmark models in the M4-Competition dataset, with significant improvements upon strong benchmark models and stable performance across the main time frequencies.

5.5. Prediction intervals

So far, the predictive properties of the structural Theta model have been evaluated in the M4 dataset, focusing on point forecasts. This section turns to prediction intervals and probabilistic performance metrics.

Table 7 examines the performance of the structural Theta model with respect to the average Mean Scaled Interval Score (MSIS) introduced by Gneiting and Raftery (2007), which has also been employed in the M4-Competition for ranking interval forecasts. The benchmark methods used for comparison are Naive, SES, Holt, Damped, DOTM and Auto Theta.

The results demonstrate that the structural Theta model significantly outperforms Naive, SES, and Holt benchmarks across yearly, quarterly, and monthly frequencies. However, Auto Theta is the most accurate method at the yearly frequency, outperforming the structural Theta. For quarterly data, Damped ranks first, followed closely by Auto Theta and the structural Theta. Conversely, the structural Theta model exhibits the highest accuracy at the monthly frequency, with DOTM and Auto Theta following closely behind.

Overall, we conclude that the structural Theta model produces accurate forecasts not only in terms of point forecasts but also in terms of prediction intervals, despite some signs of underestimation of uncertainty, particularly when forecasting yearly time series.

Table 7
Forecasting performance in terms of precision of prediction intervals across the Yearly, Quarterly and Monthly M4-Competition series.

	Yearly	Quarterly	Monthly
Naive	56.554	14.073	12.300
SES	56.038	13.595	10.923
Holt	47.427	11.833	11.385
Damped	43.419	**11.258**	10.439
MSOE Theta	46.777	11.741	**9.576**
DOTM	48.847	12.255	9.835
Auto Theta	**33.127**	11.464	9.830

Notes. The performance of the prediction intervals is evaluated using MSIS. For each frequency, the most accurate method for MSIS is highlighted in boldface. A 95% prediction interval framework for estimating the uncertainty around the point forecasts has been adopted.

Table 8
Forecasting performance of the structural Theta in the M3-competition using SMAPE.

	Average of forecasting horizon					
	1–4	1–6	1–8	1–12	1–15	1–18
Yearly	13.17	15.80				
Quarterly	7.11	8.11	8.94			
Monthly	12.49	12.38	12.74	13.11	13.72	14.42
Other	4.44	5.58	6.43			
All data	10.82	11.65	11.86	12.26	12.77	13.37

6. Empirical evaluation using the M3-Competition data

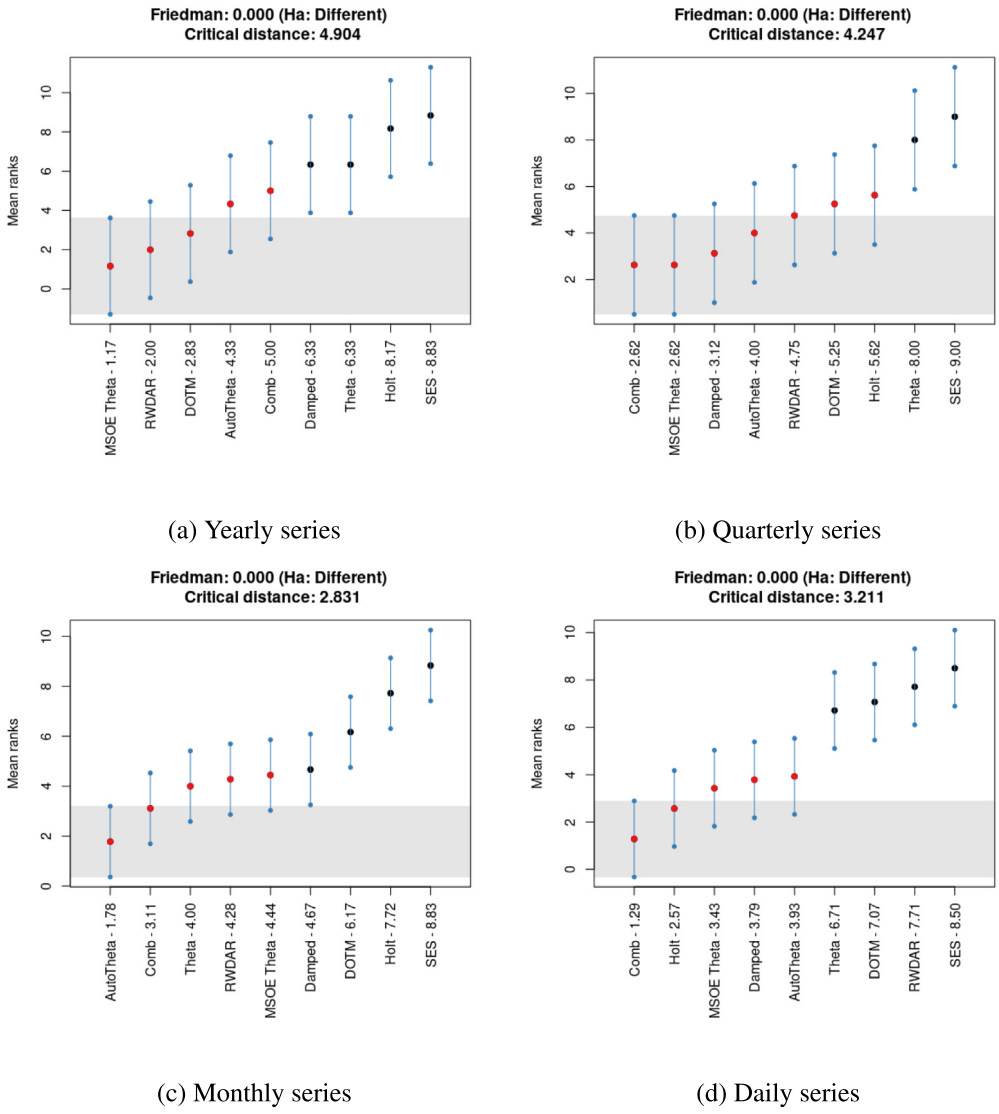
This section evaluates the performance of the structural Theta model using the M3-Competition dataset. Although the M3-Competition considers only 3003-time series, it is especially relevant in our framework since it first tested and discovered the strong potential of the classic Theta model as proposed by Assimakopoulos and Nikolopoulos (2000). Indeed, using the words of the M3 competition organisers (Makridakis & Hibon, 2000): *"[...] a new method, Theta, seems to perform extremely well. Although this method seems simple to use [...] it performs remarkably well across different types of series, forecasting horizons, and accuracy measures."*.

The M3-Competition includes various types of data (micro, industry, macro, finance, demographic, other) and different time intervals (yearly, quarterly, monthly, other). To compare the performance of the structural Theta model with the M3 competitors, we computed the SMAPE. Our results, reported in Table 8, can be compared with those of Makridakis and Hibon (2000) (see their Tables 6, 13, 14, 15, 16).

Results show that the structural Theta performs better than the classic Theta for yearly data, similarly to the classic Theta for quarterly data, and worse for monthly and other data. Overall, using the average horizons, the structural Theta performs slightly worse than the classic Theta but still ranks second or third among all competing methods.

7. Empirical evaluation using the M5-Competition data

One could raise concerns about whether the excellent forecasting performance of the structural Theta model is solely attributable to its application in the M4-competition



(a) Yearly series

(b) Quarterly series

(c) Monthly series

(d) Daily series

Fig. 3. Comparison of forecast models using multiple comparisons with the best method.
Notes. This figure shows the Nemenyi test using MASE as error measure for yearly, quarterly, monthly and daily series. The confidence level used for the comparison is 5%. The models are ordered based on their average ranks.

dataset. To address this issue, we evaluated its prediction performance in the M5-competition, which followed the M4-competition and exclusively focused on retail sales forecasting. The M5 dataset contains 42,840 hierarchical sales records from Walmart, covering stores in three US states (California, Texas, and Wisconsin). It includes item level, department, product category, and store details for a period of 5 years, starting from January 29, 2011, to April 24, 2016. The goal of the M5 competition is to forecast daily sales for the next 28 days until May 22, 2016; see Makridakis et al. (2022) for additional details. The hierarchical aggregation structure of the dataset makes it feasible to produce forecasts using a bottom-up approach or a top-down approach.

One of the main features of the M5-competition dataset is the widespread presence of zeros. This presents a challenge for standard time-series models, as they can hardly accommodate the presence of many zeros in their specifications. Despite this challenge, Table 9 reports the performance achieved by the structural Theta model in terms of point forecasts, as measured by both the overall accuracy and across the 12 aggregation levels, using the Weighted Root Mean Squared Scaled Error (WRMSSE) metrics. Table 9 also presents the performance of the structural Theta model in terms of prediction intervals based on the Weighted Scaled Pinball Loss (WSPL) metrics.

Regarding point forecasts, the structural Theta model provides accurate predictions when compared to other standard benchmarks, such as Naive, seasonal Naive, simple exponential smoothing (SES), ARIMA, ARIMAX,

Table 9
Forecasting performance of the structural Theta in the M5-competition in terms of WRMSSE (accuracy) and WSPL (uncertainty).

	Aggregation levels												Average
	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	
	Accuracy competition (WRMMSE)												
TD	0.442	0.538	0.609	0.478	0.562	0.603	0.700	0.665	0.753	1.043	0.984	0.916	0.691
BU	0.443	0.531	0.597	0.489	0.564	0.590	0.662	0.656	0.736	1.016	0.971	0.916	0.681
	Uncertainty competition (WSPL)												
TD	0.144	0.149	0.168	0.151	0.173	0.166	0.192	0.181	0.207	0.314	0.297	0.302	0.203

Notes. The results are presented at the aggregation level, as described in Makridakis et al. (2022), and overall. "Average" is the average of all aggregation levels.

etc.[12] On average, the top-down (TD) structural Theta ranks fifth among the 24 benchmark methods, while the bottom-up (BU) structural Theta ranks third, even outperforming ARIMAX models and combinations. This is not surprising given that the best-performing benchmark (Exponential Smoothing bottom-up) is a similar method, except for the presence of a drift.

However, the most relevant results pertain to the uncertainty competition (WSPL measure). Our approach outperforms all six benchmarks, including Naive and seasonal Naive, ETS, SES, ARIMA and empirical quantile estimation (Kernel).

While these results are promising, they should be interpreted cautiously as the structural Theta model is not primarily designed to handle intermittent data. Nevertheless, they demonstrate that the structural Theta model can perform reasonably well even when applied to retail sales from the M5-Competition dataset, which exhibits intermittency.

8. Conclusions

The Theta is a well-established forecasting method that has found widespread application in forecast competitions over the past two decades. This paper discusses the MSOE version of the Theta method in its state-space form, belonging to the family of structural time series models. The proposed model corresponds to a random walk plus constant with an additional error term, and its forecast function is equal to a random walk update plus a constant linear trend. An extensive examination of its out-of-sample forecasting capabilities has been conducted using the M4 competition database, and has also been performed on the M3 competition and M5 competition datasets.

Results highlight that the structural Theta model provides a simple yet effective approach to forecasting, especially when the data exhibit persistence and drift components, as confirmed by simulations. Indeed, using the M4-dataset, this model consistently outperforms all the Theta variants and other competitors across 100,000-time series. The structural Theta scores well also in the M3-Competition dataset, in which the classic Theta performed particularly strongly. Even when evaluated on the M5-competition data, which consists of hierarchical sales data known for its intermittent and unpredictable patterns, the structural Theta model exhibits remarkably strong forecasting performance. Thus, given its notable forecasting accuracy and robust performance in M3, M4 and M5, the structural Theta emerges as a highly suitable model and a reference point to benchmark in forecast competitions. Another merit of the structural Theta model is that it is fully replicable and can be estimated with minimal computational effort. In light of this, we believe this particular variant of the Theta model holds the potential for valuable application in economic and business forecasting.

[12] We refer to the complete set of results and benchmarks used in the M5-Competition as reported at https://github.com/Mcompetitions/M5-methods

Disclaimer

The views expressed herein are those of the authors and do not necessarily reflect those of the Bank of Italy – Financial Intelligence Unit or the Eurosystem.

CRediT authorship contribution statement

Giacomo Sbrana: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andrea Silvestrini:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The R codes used to obtain the results as in Table 3 to 9 can be downloaded using the following link: https://github.com/giacsbrana/Structural-THETA-Codes-and-results. In addition, the M5 dataset to reproduce the results can be downloaded using the following link: https://github.com/giacsbrana/M5_dataset. For further inquiries please contact the corresponding author.

References

Andrews, Donald W. K., & Chen, Hong-Yuan (1994). Approximately median-unbiased estimation of autoregressive models. *Journal of Business & Economic Statistics*, *12*(2), 187–204.

Assimakopoulos, Vassilis, & Nikolopoulos, Konstantinos (2000). The theta model: a decomposition approach to forecasting. *International Journal of Forecasting*, *16*(4), 521–530.

Casals, José, Sotoca, Sonia, & Jerez, Miguel (2015). Single and multiple error state-space models for signal extraction. *Journal of Statistical Computation and Simulation*, *85*(5), 1053–1069.

Castle, Jennifer L., Doornik, Jurgen A., & Hendry, David F. (2021). Forecasting principles from experience with forecasting competitions. *Forecasting*, *3*(1), 138–165.

Demšar, Janez (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, *7*(1), 1–30.

Dickey, David A., & Fuller, Wayne A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, *74*(366a), 427–431.

Dickey, David, & Fuller, Wayne A. (1981). Likelihood ratio statistics for autoregressive time series with a unit root. *Econometrica*, *49*(4), 1057–1072.

Enders, W. (1995). *Wiley series in probability and statistics - applied probability and statistics section series*, *Applied econometric times series*. Wiley.

Fiorucci, Jose A., Pellegrini, Tiago R., Louzada, Francisco, Petropoulos, Fotios, & Koehler, Anne B. (2016). Models for optimising the theta method and their relationship to state space models. *International Journal of Forecasting*, *32*(4), 1151–1161.

Gneiting, Tilmann, & Raftery, Adrian E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, *102*(477), 359–378.

Harvey, Andrew C. (1989). *Forecasting, structural time series analysis, and the Kalman filter*. Cambridge.

Hollander, M., Wolfe, D. A., & Chicken, E. (2013). *Wiley series in probability and statistics*, *Nonparametric statistical methods*. Wiley.

Hyndman, Rob J., & Billah, Baki (2003). Unmasking the Theta method. *International Journal of Forecasting*, *19*(2), 287–290.

Hyndman, Rob, & Khandakar, Yeasmin (2008). Automatic time series forecasting: The forecast package for R. *Journal of Statistical Software*, *27*(03), 1–22.

Koning, Alex J., Franses, Philip Hans, Hibon, Michele, & Stekler, H. O. (2005). The M3 competition: Statistical tests of the results. *International Journal of Forecasting*, *21*(3), 397–409.

Makridakis, Spyros (1993). Accuracy measures: theoretical and practical concerns. *International Journal of Forecasting*, *9*(4), 527–529.

Makridakis, Spyros, & Hibon, Michèle (2000). The M3-Competition: results, conclusions and implications. *International Journal of Forecasting*, *16*(4), 451–476, The M3-Competition.

Makridakis, Spyros, Spiliotis, Evangelos, & Assimakopoulos, Vassilios (2020a). The M4 competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting*, *36*(1), 54–74.

Makridakis, Spyros, Spiliotis, Evangelos, & Assimakopoulos, Vassilios (2020b). Predicting/hypothesizing the findings of the M4 competition. *International Journal of Forecasting*, *36*(1), 29–36, M4 Competition.

Makridakis, Spyros, Spiliotis, Evangelos, & Assimakopoulos, Vassilios (2022). The M5 competition: Background, organization, and implementation. *International Journal of Forecasting*, *38*(4), 1325–1336.

Nelson, Charles R., & Plosser, Charles I. (1982). Trends and random walks in macroeconmic time series: Some evidence and implications. *Journal of Monetary Economics*, *10*(2), 139–162.

Nemenyi, Peter Bjorn (1963). *Distribution-free multiple comparisons*. Princeton University.

Nikolopoulos, Kostas I., & Thomakos, Dimitrios D. (2019). Forecasting with the Theta method: Theory and applications. In *Forecasting with the Theta method*. John Wiley and Sons, Ltd.

Oh, Kum Hwa, Zivot, Eric, & Creal, Drew (2008). The relationship between the Beveridge-Nelson decomposition and other permanent-transitory decompositions that are popular in economics. *Journal of Econometrics*, *146*(2), 207–219.

Rossana, Robert, & Seater, John (1995). Temporal aggregation and economic time series. *Journal of Business & Economic Statistics*, *13*(4), 441–451.

Sbrana, Giacomo, & Silvestrini, Andrea (2023). The RWDAR model: A novel state-space approach to forecasting. *International Journal of Forecasting*, [ISSN: 0169-2070] *39*(2), 922–937.

Spiliotis, Evangelos, Assimakopoulos, Vassilios, & Makridakis, Spyros (2020). Generalizing the Theta method for automatic forecasting. *European Journal of Operational Research*, *284*(2), 550–558.

Spiliotis, Evangelos, Assimakopoulos, Vassilios, & Nikolopoulos, Konstantinos (2019). Forecasting with a hybrid method utilizing data smoothing, a variation of the Theta method and shrinkage of seasonal factors. *International Journal of Production Economics*, *209*, 92–102, The Proceedings of the 19th International Symposium on Inventories.

Spiliotis, Evangelos, Kouloumos, Andreas, Assimakopoulos, Vassilios, & Makridakis, Spyros (2020). Are forecasting competitions data representative of the reality? *International Journal of Forecasting*, *36*(1), 37–53, M4 Competition.

Thomakos, Dimitrios D., & Nikolopoulos, Konstantinos (2015). Forecasting multivariate time series with the theta method. *Journal of Forecasting*, *34*(3), 220–229.

Watson, Mark W. (1986). Univariate detrending methods with stochastic trends. *Journal of Monetary Economics*, *18*(1), 49–75.

Working, Holbrook (1960). Note on the correlation of first differences of averages in a random chain. *Econometrica*, *28*(4), 916–918.